

Design of a dq-axis Current Feedback Controller with Two Inputs of Voltage and Frequency for S-S Type Inductive Power Transfer System

Abstract—Inductive Power Transfer (IPT) is attracting attention as a key technology for resolving charging issues in Electric Vehicle. In IPT systems, fast and precise current control is essential to ensure stable and efficient power transmission. To achieve this challenge, this paper proposes a current control strategy based on the derivation of a dynamic model with PWM-controlled inverter output voltage and operating frequency and a controller design using the pole placement method. The validity of the proposed control system is verified through experimental step response tests using a digital control platform. Experimental results demonstrate that the dq-axis currents converge to their reference values within approximately 20 ms and 60 ms, respectively, achieving zero steady-state error. Although transient overshoots attributed to cross-coupling effects neglected in the Single-Input Single-Output (SISO) design were observed, the study confirms the system's stability. It is concluded that while the proposed method is effective, a Multi-Input Multi-Output (MIMO) design is essential for further suppressing interference and improving transient performance.

Index Terms—Inductive Power Transfer, Design of Controller, Feedback Control, dq-axis Current, Frequency Control

I. INTRODUCTION

Inductive power transfer (IPT) systems have attracted considerable attention as a promising technology for addressing challenges in electric vehicle charging [1]–[3]. However, prior to widespread practical application, a variety of issues must be addressed. These range from system-performance challenges, such as reduced transmission efficiency and instability under load variations, to societal concerns, including road surface materials and safety regarding magnetic field leakage [4], [5]. While social implementation remains a key challenge, achieving high technical performance is equally critical for practical realization.

Previous studies have proposed various approaches to address these performance issues. These include hardware improvements, such as optimized coil designs and circuit configurations, as well as control-oriented enhancements like inverter switching strategies [6].

In particular, control-focused research has investigated voltage amplitude control via Pulse Width Modulation (PWM) control, operating frequency control, or a combination of both [7]–[10]. Simultaneously controlling voltage amplitude and operating frequency offers greater flexibility, which is advantageous for efficiency optimization and robustness against load variations. Nevertheless, in many previous studies, frequency control has been utilized primarily to achieve zero-voltage switching (ZVS), and the full potential of two-input control

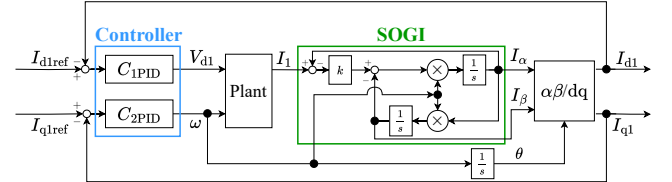


Fig. 1. Block Diagram of Proposed Controller Design

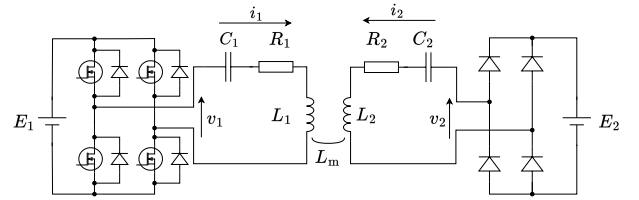


Fig. 2. Circuit diagram of the S-S IPT system

has not yet been fully exploited. In [11], a model and controller design utilizing two inputs, voltage and frequency, were developed. Although this method successfully reduced the system to five states while maintaining sufficient accuracy, it assumes a constant resistive load, and the frequency input is dedicated exclusively to achieving ZVS.

In this study, a two-input model incorporating voltage and frequency is derived under a constant voltage load condition, which better represents battery charging. Furthermore, a controller is designed in which the operating frequency is dynamically regulated to improve performance, rather than being constrained solely to maintaining ZVS. The proposed approach is experimentally validated, and its advantages and disadvantages are discussed.

II. DERIVATION OF A TWO-INPUT MODEL AND CONTROLLER DESIGN

Fig. 1 shows the block diagram of the proposed controller design, while Fig. 2 illustrates the series-series (SS) compensated circuit used as the plant. The system consists of a dq transformation synchronized with the phase of the fundamental inverter output voltage v_1 using a Second-Order Generalized Integrator (SOGI) [12].

Inspired by the P/ω and Q/V droop control strategies widely established in power grid systems [13], this paper applies an analogous approach to the IPT system. In this

context, the q-axis current corresponds to reactive power, while the d-axis current corresponds to active power. Leveraging the characteristic that the polarity of the q-axis current I_{q1} is determined by the operating angular frequency ω relative to the resonant frequency, and the magnitude of the d-axis current I_{d1} is controlled by the PWM-controlled inverter output voltage V_{d1} , the proposed controller regulates I_{q1} using ω and I_{d1} using V .

Here, $j = d$, $j = q$ correspond to the dq-axes, respectively, and the controller is expressed by the following equations using the pseudo-derivative filter coefficient N_j .

$$C_{PIDj} = K_{pj} + \frac{K_{ij}}{s} + \frac{K_{dj}N_j}{1 + N_j \frac{1}{s}} \quad (1)$$

A. Derivation of a Two-Input Model

Based on Kirchhoff's laws, the following fundamental-component voltage equations are obtained.

$$v_1 = R_1 i_1 + v_{C1} + L_1 \frac{di_1}{dt} + L_m \frac{di_2}{dt} \quad (2)$$

$$v_2 = R_2 i_2 + v_{C2} + L_2 \frac{di_2}{dt} + L_m \frac{di_1}{dt} \quad (3)$$

Here, by treating the operating angular frequency ω as a time-varying parameter $\omega(t)$, the variables v_1 , v_2 , i_1 , i_2 , v_{C1} and v_{C2} in dq reference frame are expressed as follows.

$$\omega = \omega(t) \quad (4)$$

$$v_i = V_{di} \cos\{\omega(t)t\} - V_{qi} \sin\{\omega(t)t\} \quad (5)$$

$$i_i = I_{di} \cos\{\omega(t)t\} - I_{qi} \sin\{\omega(t)t\} \quad (6)$$

$$v_{Ci} = V_{Cdi} \cos\{\omega(t)t\} - V_{Cqi} \sin\{\omega(t)t\} \quad (i \in \{1, 2\}) \quad (7)$$

The state vector $\mathbf{x}$ and the input vector $\mathbf{u}$ are defined as shown below. In this study, since the rotating reference frame is aligned with the voltage vector v_1 , the condition $V_{q1} = 0$ is satisfied. Thus, the input vector $\mathbf{u}$ consists of the d-axis voltage V_{d1} and angular frequency ω . By rearranging the circuit equations using the mutual inductance $L_m = k\sqrt{L_1 L_2}$, the state-space equation is derived as (10). Due to page constraints, the detailed derivation is omitted.

$$\mathbf{x} = (I_{d1} \ I_{q1} \ I_{d2} \ I_{q2} \ V_{cd1} \ V_{cq1} \ V_{cd2} \ V_{cq2})^T \quad (8)$$

$$\mathbf{u} = (V_{d1} \ \omega(t))^T \quad (9)$$

Eq. (10) contains multiple nonlinearities arising from two sources: the time derivative of ω and the product terms of the state and input variables.

First, since the time variation of the operating angular frequency is sufficiently smaller than the operating angular frequency itself, its time derivative is approximated as zero as shown in the following equation. This simplification eliminates the nonlinearity associated with the frequency variation.

$$\frac{d\omega(t)}{dt} \simeq 0 \quad \left(\cdot \cdot \frac{d\omega(t)}{dt} \ll \omega(t) \right) \quad (11)$$

TABLE I
CIRCUIT PARAMETER

Symbol	Value	Symbol	Value
R_1	1.26 Ω	R_2	0.956 Ω
L_1	549.13 μH	L_2	433.76 μH
C_1	6.28 nF	C_2	7.98 nF
L_m	48.61 μH	k	0.0996
E_1	30 V	E_2	10 V
V_{d10}	25 V	f_0	81.5 kHz

Next, to address the remaining nonlinearities, the system is linearized around an equilibrium point defined by the state vector $\mathbf{x}_0$ and the input vector $\mathbf{u}_0$. Here, $\mathbf{u}_0$ can be arbitrarily chosen to satisfy the equilibrium condition.

$$\mathbf{x}_0 = (I_{d10} \ I_{q10} \ I_{d20} \ I_{q20} \ V_{cd10} \ V_{cq10} \ V_{cd20} \ V_{cq20})^T \quad (12)$$

$$\mathbf{u}_0 = (V_{d10} \ \omega_0)^T \quad (13)$$

The coefficients of the linearized state equation can be obtained from the following equations, and the corresponding transfer function can be derived from the state-space representation. Using these coefficients, the transfer function is derived by (16). In the following discussion, Δ is omitted for simplicity.

$$\Delta \mathbf{A} = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_{\mathbf{x}=\mathbf{x}_0, \mathbf{u}=\mathbf{u}_0}, \quad \Delta \mathbf{B} = \left. \frac{\partial f}{\partial \mathbf{u}} \right|_{\mathbf{x}=\mathbf{x}_0, \mathbf{u}=\mathbf{u}_0} \quad (14)$$

$$\Delta \mathbf{C} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}, \quad \Delta \mathbf{D} = \mathbf{0} \quad (15)$$

$$\Delta \mathbf{P} = \Delta \mathbf{C}(s\mathbf{I} - \mathbf{A})^{-1} \Delta \mathbf{B} + \Delta \mathbf{D} \quad (16)$$

Hereafter, the transfer function is represented by (17). This transfer function serves as the basis for the controller design discussed in the next section.

$$\mathbf{P}(s) = \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} = \begin{pmatrix} \frac{I_{d1}}{V_{d1}} & \frac{I_{d1}}{\omega(t)} \\ \frac{I_{q1}}{V_{d1}} & \frac{I_{q1}}{\omega(t)} \end{pmatrix} \quad (17)$$

B. Model Order Reduction Based on Bode Plots

TABLE I summarizes the circuit parameters used in this study. In this paper, although the model input is defined as the angular frequency ω , the frequency f ($\omega = 2\pi f$) is used to describe the equilibrium point and experimental conditions for physical clarity. As noted earlier, there is a degree of freedom in selecting the input equilibrium values V_{d10} and f_0 . Although the choice of the equilibrium point may affect the model accuracy, its influence is beyond the scope of this paper. Therefore, an equilibrium point suitable for the frequency characteristics of the current is selected in this study. The Bode

$$\frac{d}{dt} \mathbf{x} = \mathbf{f}(\mathbf{x}, \mathbf{u}) = \begin{pmatrix} \frac{R_1}{(k^2-1)L_1} I_{d1} + \left(\frac{d\omega}{dt} t + \omega\right) I_{q1} - \frac{kR_2}{(k^2-1)\sqrt{L_1 L_2}} I_{d2} + \frac{1}{(k^2-1)L_1} V_{cd1} - \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{cd2} - \frac{1}{(k^2-1)L_1} V_{d1} + \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{d2} \\ - \left(\frac{d\omega}{dt} t + \omega\right) I_{d1} + \frac{R_1}{(k^2-1)L_1} I_{q1} - \frac{kR_2}{(k^2-1)\sqrt{L_1 L_2}} I_{q2} + \frac{1}{(k^2-1)L_1} V_{cq1} - \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{cq2} - \frac{1}{(k^2-1)L_1} V_{q1} + \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{q2} \\ - \frac{kR_1}{(k^2-1)\sqrt{L_1 L_2}} I_{d1} + \frac{R_2}{(k^2-1)L_2} I_{d2} + \left(\frac{d\omega}{dt} t + \omega\right) I_{q2} - \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{cd1} + \frac{1}{(k^2-1)L_2} V_{cd2} + \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{d1} - \frac{1}{(k^2-1)L_2} V_{d2} \\ - \frac{kR_1}{(k^2-1)\sqrt{L_1 L_2}} I_{q1} - \left(\frac{d\omega}{dt} t + \omega\right) I_{d2} + \frac{R_2}{(k^2-1)L_2} I_{q2} - \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{cq1} + \frac{1}{(k^2-1)L_2} V_{cq2} + \frac{k}{(k^2-1)\sqrt{L_1 L_2}} V_{q1} - \frac{1}{(k^2-1)L_2} V_{q2} \\ \frac{1}{C_1} I_{d1} + \left(\frac{d\omega}{dt} t + \omega\right) V_{cq1} \\ \frac{1}{C_1} I_{q1} - \left(\frac{d\omega}{dt} t + \omega\right) V_{cd1} \\ \frac{1}{C_2} I_{d2} + \left(\frac{d\omega}{dt} t + \omega\right) V_{cq2} \\ \frac{1}{C_2} I_{q2} - \left(\frac{d\omega}{dt} t + \omega\right) V_{cd2} \end{pmatrix} \quad (10)$$

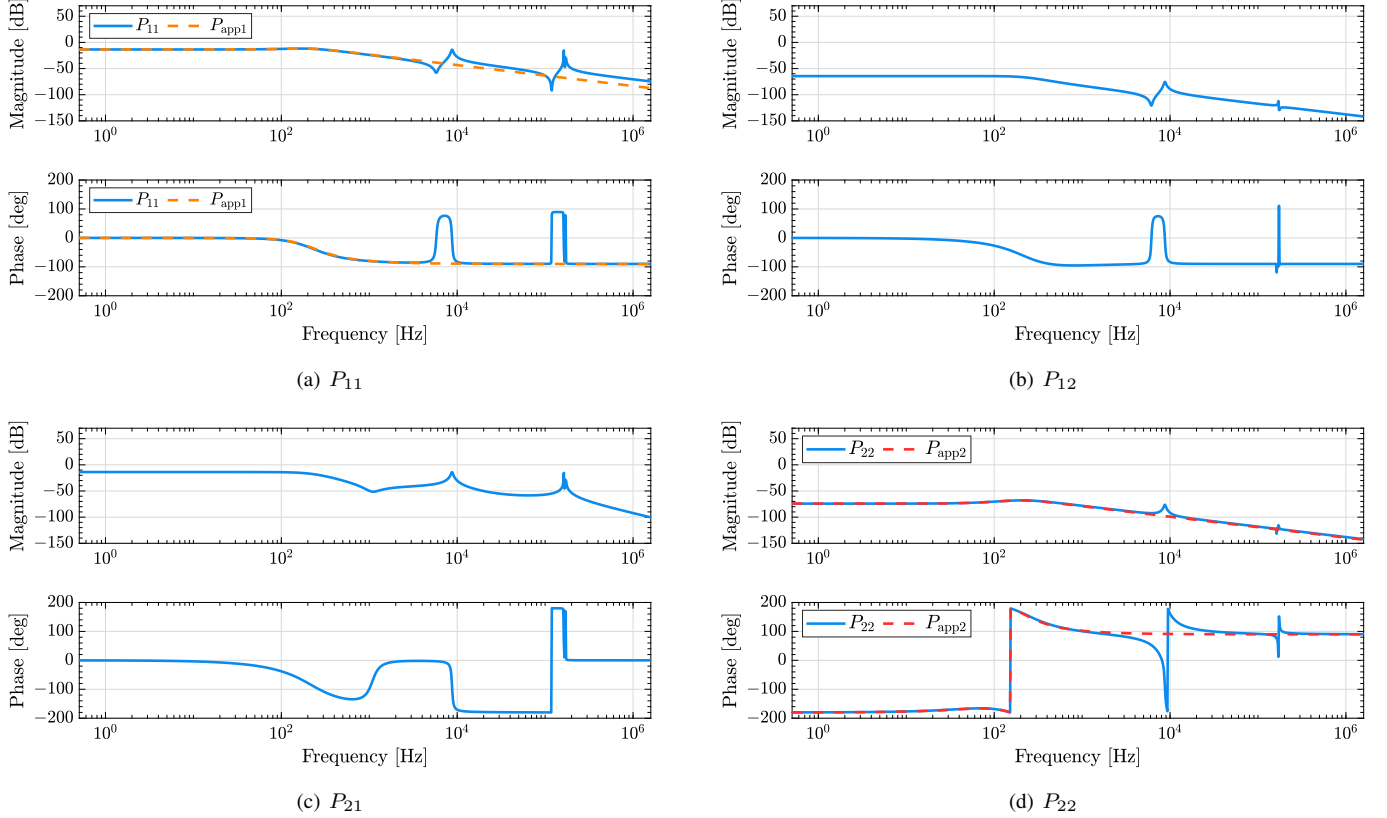


Fig. 3. Bode Diagrams of Original and Reduced-Order Models

plots of both the original and reduced-order models are shown in Fig. 3. It is observed that the reduced-order model provides a good approximation up to 2000 Hz for P_{11} and 200 Hz for P_{22} .

C. Controller Design Based on Pole Placement

Analogous to the control strategies in grid-connected inverters, this study designs a controller where I_{d1} is regulated by V_{d1} , and I_{q1} is regulated by ω .

The closed-loop poles are selected within the frequency ranges where the reduced-order approximation remains valid, specifically, below 2000 Hz for P_{11} and below 200 Hz for P_{22} . In this study, the poles are assigned with a multiplicity of

TABLE II
GAIN BY POLE PLACEMENT

Symbol	Value	Symbol	Value
K_{pd}	2.1	K_{pq}	2502.5
K_{id}	5501.3	K_{iq}	-2.2×10^6
K_{dd}	0.0021	K_{dq}	-7.1
N_d	1027.2	N_q	531.0

four at 200 Hz for the P_{11} loop and 130 Hz for the P_{22} loop. TABLE II shows the obtained gains.

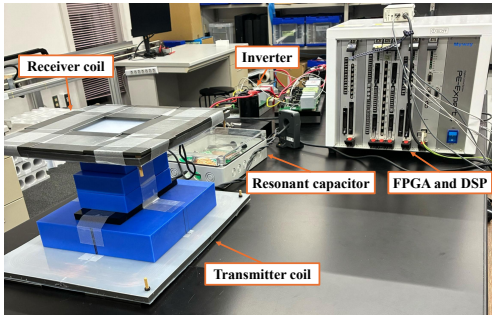


Fig. 4. Experimental Environment

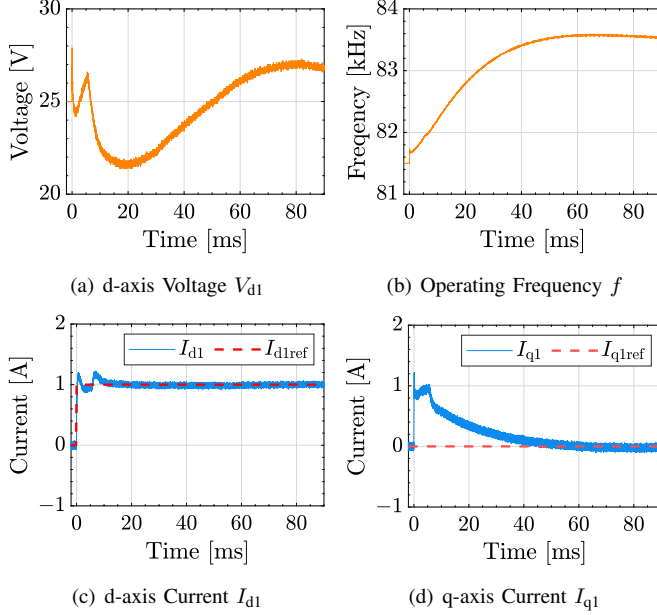


Fig. 5. Experimental Results of The Step Response

III. EXPERIMENTAL RESULTS

The experimental setup is shown in Fig. 4. The control system is implemented using a hybrid architecture consisting of a Field Programmable Gate Array (FPGA) and a Digital Signal Processor (DSP). High-speed signal processing tasks, such as the SOGI-based filtering and dq coordinate transformation, are executed on the FPGA to minimize detection latency. Meanwhile, the DSP handles the calculation of the control inputs based on the proposed pole placement law. The circuit parameters are listed in TABLE I, which are identical to those used in the previous section.

To evaluate the dynamic performance and validity of the proposed controller, a step response test was conducted under a constant voltage load and mutual inductance condition. The d-axis current reference I_{d1ref} was stepped up from 0 A to 1 A, while the q-axis current reference I_{q1ref} was maintained at 0 A. The DC power supply voltage E_1 was set to 30 V, and the battery voltage was simulated using a constant voltage load E_2 of 10 V. The control periods for V_{d1} and f were set to 15 μ s and 30 μ s, respectively.

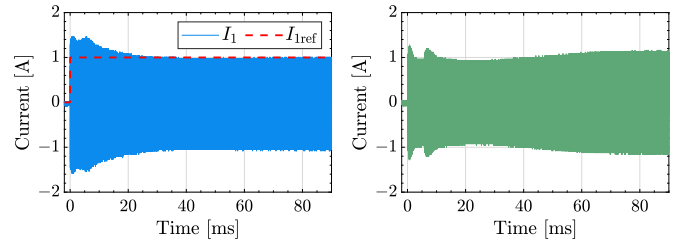


Fig. 6. Oscilloscope Capture of Current Waveforms

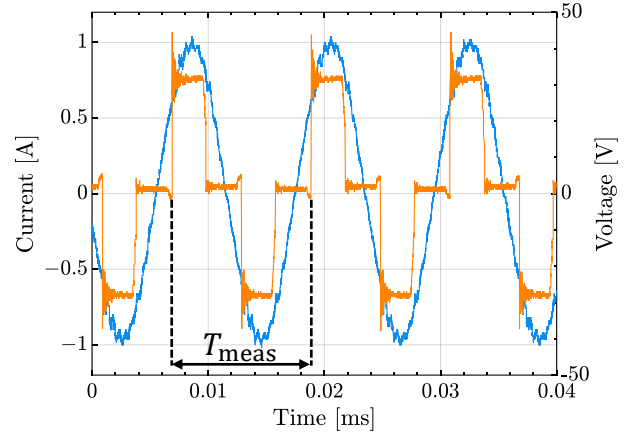


Fig. 7. Steady State Experimental Waveforms of v_1 and i_1

A. Step Response of dq-Axis Currents

Fig. 5 shows the experimental results of the step response. The control inputs, namely V_{d1} and f , are shown in Figs. 5 (a) and (b), respectively. The operating frequency in the steady state f_{steady} was observed to be 83.56 kHz. This value is utilized in the later discussion. The resulting I_{d1} and I_{q1} are depicted in Figs. 5 (c) and (d). As observed in Fig. 5 (c), I_{d1} converges to the reference value within approximately 20 ms, whereas I_{q1} in Fig. 5 (d) settles in roughly 60 ms. Therefore, the steady-state error becomes zero in both axes, confirming the effectiveness of the integral action. However, note that overshoots are observed in both waveforms during the transient state.

Notably, it is observed that V_{d1} continues to vary even after I_{d1} has converged at approximately 20 ms. This voltage variation appears to be synchronized with the changes in f , indicating that V_{d1} is fluctuating due to interference from the frequency loop even after the current has settled. This observation underscores the significant impact of the cross-coupling effects that were neglected during the design phase. Moreover, the large variation in frequency indicates that the system state deviates from the nominal equilibrium point used for linearization. This deviation introduces model mismatch due to the inherent nonlinearity of the WPT circuit. Con-

sequently, it is inferred that the observed current overshoot is induced by the combined effects of the cross-coupling and the parameter discrepancy caused by this operating point shift. In particular, given that the impact of cross-coupling is considered dominant, introducing a decoupling control scheme to compensate for these off-diagonal terms is expected to be effective in further suppressing the overshoot and improving the transient response.

B. Time-Domain Voltage and Current Waveforms

Figs. 6 (a) and (b) present the oscilloscope waveforms of the transmitter and receiver currents respectively to further analyze the tracking performance. From Fig. 6 (a), it can be seen that the currents converge at the same time as the previously discussed dq-axis currents. This consistency verifies that the dq transformation in this study is performed correctly. Furthermore, Fig. 6 (b) confirms the flow of the receiver current. This demonstrates that the proposed method maintains controllability during receiver conduction.

Notably, although I_1 settles at a constant value, I_2 exhibits an increasing trend. This phenomenon stems from the increase in V_1 , dictated by the nature of the WPT circuit topology. Consequently, controlling I_2 is considered an essential task for future system implementation.

To further verify the correct implementation of the proposed method, the discussion proceeds using the oscilloscope waveforms of the voltage and current shown in Fig. 7. As previously noted, f_{steady} was 83.56 kHz. On the other hand, the measured period T_{meas} obtained from Fig. 7 is 11.96 μs , which corresponds to a frequency of 83.61 kHz. This error originates from the sampling limitation of the oscilloscope and the digital resolution. Despite this, the close correspondence between the command reference and the actual frequency verifies the effectiveness of the dynamic frequency control.

IV. CONCLUSION

In this paper, a current control system based on the pole placement method was proposed and experimentally verified. First, a linearized state-space model was derived, explicitly treating the d-axis voltage and operating frequency as two independent control inputs. Drawing inspiration from control strategies used in grid-connected inverters, a feedback controller was designed to regulate the currents based on this structure. Experimental results demonstrated that the proposed system successfully regulates the currents and confirms the validity of the modeling and design approach.

Experimental results from step response tests confirmed the validity of the proposed system. The dq-axis currents converged to their reference values within approximately 20 ms and 60 ms, respectively, with zero steady-state error. These results confirm that the proposed design achieves sufficient tracking performance and stability for practical applications.

However, transient overshoots were observed in the current waveforms. As discussed, this observation highlights the significant impact of cross-coupling effects between the dq-axes, which were neglected in the grid-inspired SISO-based design.

This suggests that while the proposed strategy is effective for steady-state regulation, the interference is substantial during transients. Consequently, to suppress such interference, future work will consider adopting a Multi-Input Multi-Output (MIMO) design approach, such as decoupling control, as well as implementing receiver-side control.

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